

L11. Fault-Tolerant Quantum Computation

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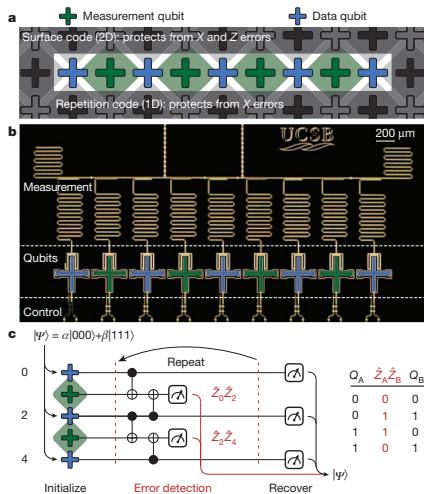
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Outline

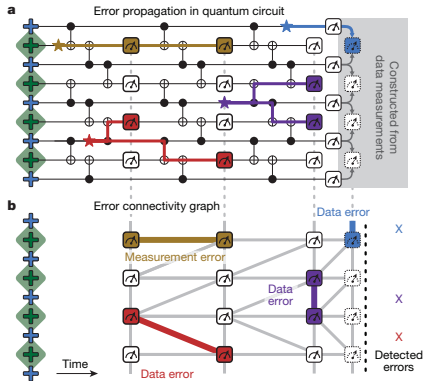
- Fault-Tolerant Quantum Computation
- Particle Statistics and Exchange
- Toric Code and Quantum Error Correction

10.1 Fault-Tolerant Quantum Computation

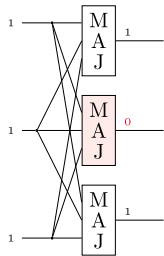
- The UCSB team reported the protection of classical states from environmental bit-flip errors [Kelly et al., Nature 519, 66 (2015)].
- Relative to a single physical qubit, the failure rate reduced by a factor of 2.7 using five qubits and by a factor of 8.5 using nine qubits.



- Data qubit errors (purple, red, blue) are detected on neighboring measurement qubits in the same or next cycle.
- An error on a measurement qubit (gold) is detected in two subsequent rounds.
- The team used minimum-weight perfect matching to decode to physical errors, based on the pattern of detection events and an error model for the system.



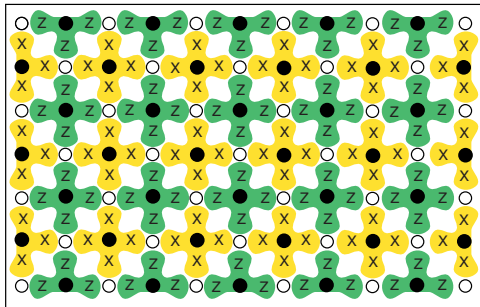
- A general notation to describe an ECC is $[n, k, d]$, where k is the number of logical bits, and d is the minimum distance between any two legal (no-error) encodings.
 - The repetition codes are $[n, 1, n]$ codes.
 - In general, a code with distance d can correct up to $(d - 1)/2$ errors.
 - With 5 bits, the code can correct up to 2 errors, because the 3 correct bits form a majority.
- We can also replicate the majority vote MAJ circuit. If one MAJ fails, however, the result is correctable by the next stage of the system.



- Therefore, even though the repetition code cannot correct all possible errors, it convinces us that noise is not a serious problem in principle; error correction can reduce the error probability from p to $O(p^2)$.
- If we further introduce hierarchical procedures, the error probability can be ultimately reduced to as low a level as desired, as long as the original error rate p is below some threshold p_{th} (with physically reasonable assumptions).
- This is known as the threshold theorem, as was also important for early-day classical computation.

Surface Code

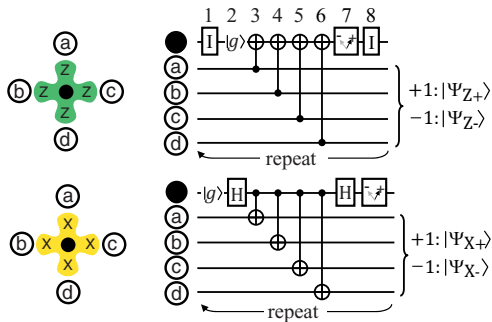
- In a surface code, qubits form a two-dimensional array. They are entangled into a randomly selected quiescent state resulting from the measurement of stabilizers.



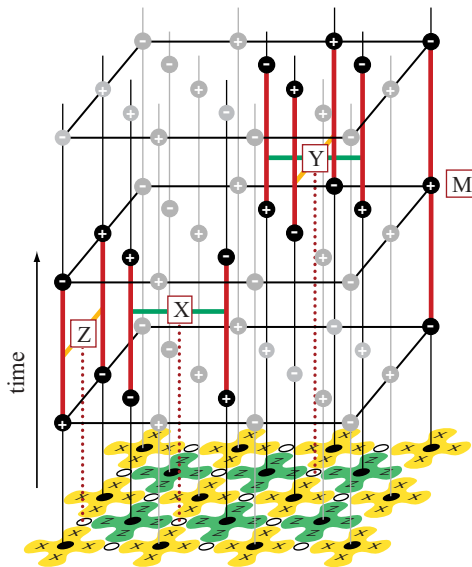
- The surface code has two families of stabilizers:

$$Z_a Z_b Z_c Z_d, \quad X_a X_b X_c X_d.$$

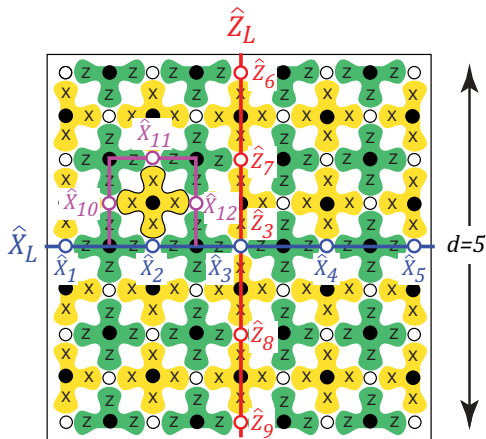
- Each stabilizer contains a sequence of CNOT operations and the subsequent measurement involving 4 data qubits and 1 measurement qubit.



- They are repeatedly measured. Qubit and measurement errors can be detected by the change of measurement results.
- If the errors are sufficiently rare, they can be undone in classical control software.



- The array of physical qubits encode one logical qubit, whose logical operations can be achieved by chains of operations on physical qubits.



10.2 Particle Statistics and Exchange

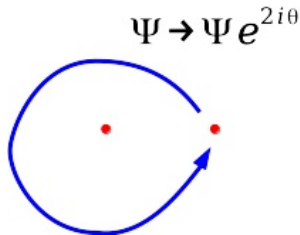
- Particle statistics is a quantum mechanical property.
 - Classical particles distinguishable with their positions being tracked.
 - Quantum particles described by a spatial wave function; impossible to define distinguishability by their positions.
- Intrinsic properties of indistinguishable particles: mass, charge, spin and any other quantum numbers.
- Change in the wave function of two identical particles, when they are exchanged. Its properties need to be compatible with the symmetry imposed by indistinguishability.

Exchange is Topological

- The effect of the particle exchange process should not depend on:
 - the particular shape of the path taken by the particles when they are exchanged, or
 - the speed the path is traversed.
- Deforming the shape of the path in a continuous way is *geometrical*. The statistical evolution is invariant over such continuous coordinate transformations. It is *topological* in nature; it only depends on some global, topological characteristics of the path, such as the number of times the particles are exchanged.

- If we perform two exchanges (accompanied by irrelevant spatial translations) in succession, we obtain a full circulation of one particle around the other:

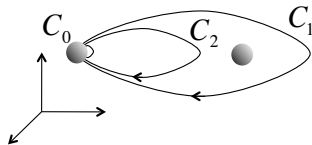
$$|\Psi(r_1, r_2)\rangle \rightarrow e^{i\theta} |\Psi(r_2, r_1)\rangle \rightarrow e^{2i\theta} |\Psi(r_1, r_2)\rangle .$$



- What are continuous deformations? What are not?

Bosons and Fermions in 3D

- Consider two identical particles in three dimensions. We can continuously deform the path C_1 to the path C_2 , or to C_0 . This involves only local deformations of the evolution without cutting or otherwise drastically changing the nature of the path at any time. So, $e^{2i\theta} = 1$, i.e., $e^{i\theta} = \pm 1$ or $\theta = 0, \pi$.



- Therefore, in 3D the indistinguishability of particles allows for the possibility of having bosons and fermions.
 - Bosons satisfy the Bose-Einstein distribution: symmetric wave functions wrt particle exchanges.

$$|\Psi(r_1, r_2)\rangle \xrightarrow{r_1 \leftrightarrow r_2} |\Psi(r_2, r_1)\rangle = |\Psi(r_1, r_2)\rangle$$

- Fermions the Fermi-Dirac distribution: completely antisymmetric wave functions wrt particle exchanges. Pauli exclusion principle.

$$|\Psi(r_1, r_2)\rangle \xrightarrow{r_1 \leftrightarrow r_2} |\Psi(r_2, r_1)\rangle = e^{i\pi} |\Psi(r_1, r_2)\rangle$$

Anyons in 2D

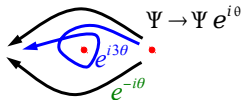
- In 2D, the particle circulation C_1 is performed on a plane, and is not possible to continuously deform it to the path C_2 , due to the existence of the other particle (i.e., a puncture).
- It is now possible to assign a phase factor to the evolution corresponding to C_1 (i.e., two successive exchanges).

$$|\Psi(r_1, r_2)\rangle \xrightarrow{r_1 \leftrightarrow r_2} |\Psi(r_2, r_1)\rangle = e^{i\theta} |\Psi(r_1, r_2)\rangle$$

These particles are known as (Abelian) anyons.

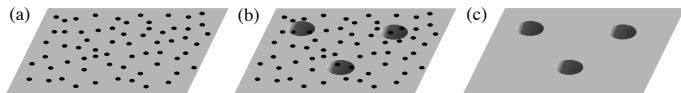
Path equiv. in 3D; NOT equiv. in 2D:

Abelian anyons (i.e., different by a phase)



Emergent Quasiparticles

- Constituent particles (e.g., electrons in a semiconductor quantum well) do not necessarily live in 2D, but their wave function is effectively 2D.
- Effective particles, or quasiparticles, emerge from the properties of 2D wave function. They behave like particles, i.e. they have local properties and they respond to their local environment.



- The constituent particles do not acquire anyonic properties. However, quasiparticles can be anyons with properties that are completely different from those of the constituent particles.
- All anyons emerge from the same wave function. They are aware of each others' position which makes it possible to exhibit the desired exchange statistics.
- More concretely, the exchange statistics emerges as an evolution operator of this wave function that depends on the history of the whole system of the constituent particles.

10.3 Toric Code and Quantum Error Correction

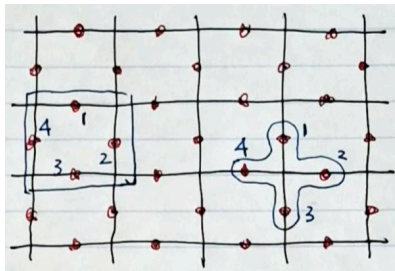
- Square lattice with qubits (or spin-1/2 particles) positioned at the lattice links (or edges).

- Vertex operator

$$A(v) = X_1 X_2 X_3 X_4$$

- Plaquette operator

$$B(p) = Z_1 Z_2 Z_3 Z_4$$



- One can recognize $A(v)$ and $B(p)$ as the stabilizers of the toric code.
 - $A^2(v) = B^2(p) = 1$.
 - All $A(v)$ and $B(p)$ commute with each other.
- Alexei Kitaev promoted the quantum error correcting code into a many-body interacting system, with a Hamiltonian

$$H = - \sum_v A(v) - \sum_p B(p),$$

whose eigenstates are also states of a quantum error correcting code and which energetically penalizes transformations between states.

Ground State

- Energy is minimized when all $A(v)$ and $B(p)$ are $+1$.

$$E_{gs} = -(\text{number of vertices}) - (\text{number of plaquettes}).$$

- The ground state wave function satisfies

$$A(v) |\psi_{gs}\rangle = |\psi_{gs}\rangle, \quad \forall v,$$

$$B(p) |\psi_{gs}\rangle = |\psi_{gs}\rangle, \quad \forall p.$$

- For example,

$$|\psi_{gs}\rangle = \prod_v \frac{1}{\sqrt{2}} [1 + A(v)] |00 \cdots 0\rangle.$$

Quasiparticle Excitations

- An elementary excitation, or a quasiparticle, occurs if one of the constraints

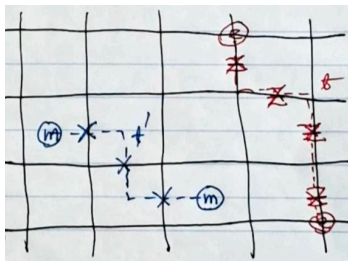
$$A(v) |\psi_{gs}\rangle = |\psi_{gs}\rangle, \quad B(p) |\psi_{gs}\rangle = |\psi_{gs}\rangle$$

is violated.

- They appear in pairs, because any X or Z appears twice in the stabilizers.
- The presence of quasiparticle excitations can be detected by measuring the eigenvalues of the corresponding $A(v)$ or $B(p)$ operators.

- Two-quasiparticle states:

$$\begin{aligned}
 |e, e\rangle &= Z |\psi_{gs}\rangle, \\
 |m, m\rangle &= X |\psi_{gs}\rangle, \\
 |\epsilon, \epsilon\rangle &= ZX |\psi_{gs}\rangle \\
 &= iY |\psi_{gs}\rangle.
 \end{aligned}$$

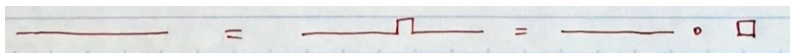


- The operators can be generalized to a string, e.g.,

$$Z(t) = \prod_{j \in t} Z_j, \quad X(t') = \prod_{j \in t'} X_j.$$

- Therefore, by applying the Pauli operators one by one we can move anyons around the lattice.

- The state of two anyons at the end of a string is invariant with respect to deformations of the shape of the string as long as its endpoints remain fixed. Hence, the exact shape of the string does not have any physical meaning. Only the position of the anyons does.

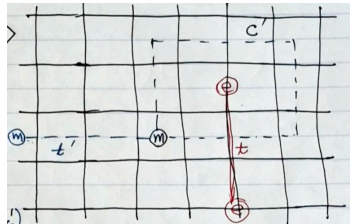


- The effect of the loop operator will be discussed soon.

Anyon Statistics

- What happens if we move one anyon around another, or exchange them up to an irrelevant translation?

$$\begin{aligned} |\psi_i\rangle &= X(t')Z(t) |\psi_{gs}\rangle, \\ |\psi_f\rangle &= X(c') |\psi_i\rangle \\ &= - |\psi_i\rangle, \end{aligned}$$

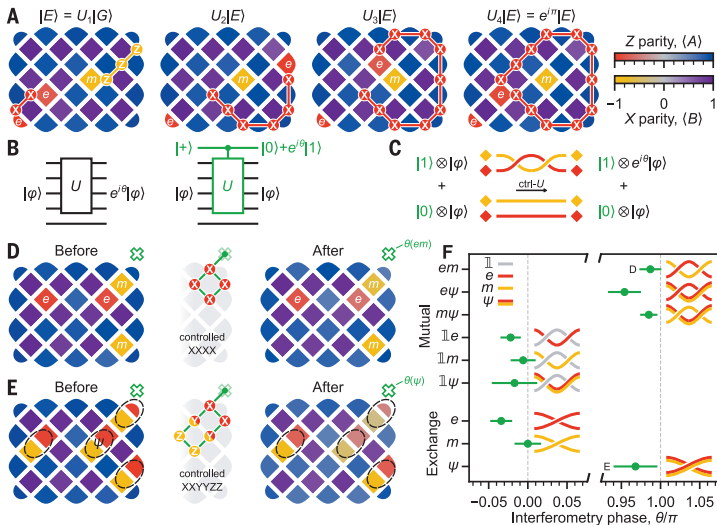


- X_j on different sites commute, so $X(c')X(t') = X(t')X(c')$.
- X_j and Z_j on the same sites anticommute, and
- X_j and $Z_{j'}$ on different sites commute, so $X(c')Z(t) = -Z(t)X(c')$.

- Neither bosons nor fermions produce such a phase change. The quasiparticles are Abelian anyons.

$$\begin{aligned}
 |\Psi(r_1, r_2; t_0)\rangle &\xrightarrow{r_1 \leftrightarrow r_2} |\Psi(r_2, r_1; t_1)\rangle = e^{i\theta} |\Psi(r_1, r_2; t_0)\rangle \\
 &\xrightarrow{r_1 \leftrightarrow r_2} |\Psi(r_1, r_2; t_2)\rangle = e^{2i\theta} |\Psi(r_1, r_2; t_0)\rangle
 \end{aligned}$$

- Note that no phase changes occur if both quasiparticles are of the same type, e.g., e around e or m around m .
- In other words, loop operators can be used to detect the enclosed anyon. This is crucial for measurement.



Simulation of
braiding
statistics with
a toric code
[Satzinger et
al., Science
374, 1237
(2021)].

Fusion of Anyons

- The ground state corresponds to the anyonic vacuum, i.e. the absence of any anyon. We denote the state by 1 .
- The outcome of combining two anyons:

$$\begin{aligned}e \times e &= m \times m = \epsilon \times \epsilon = 1, \\e \times m &= \epsilon, \quad \epsilon \times e = m, \quad \epsilon \times m = e.\end{aligned}$$

- The fusion rules specify the combination of different types of anyons. In particular, if two identical anyons are moved to the same plaquette or vertex, they annihilate.

Contracting Loops

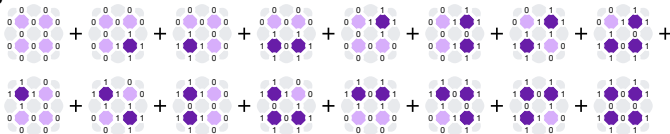
- If a string of X or Z operations forms a loop, the anyons at its ends fuse with each other (hence annihilate).
- In the discussion of anyonic statistics, we already used

$$X(c') |\psi_{gs}\rangle = |\psi_{gs}\rangle ,$$

when c' is a contracting loop.

- The ground state is the equal superposition of all possible products of elementary loops $A(v)$. Application of any contractible loop operator on the ground state gives back the same state with its components rearranged.

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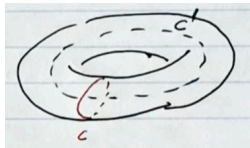
- In a dynamical view, a loop operator creates a pair of anyons (m anyons for an X loop and e for Z), move one of them along the loop, and fuse them when they meet.
 - The eigenvalue of the X loop operators detects if the loop encloses an odd or an even number of e anyons. Differently put, they read the total fusion outcome of all the enclosed e anyons.
 - Equivalent considerations hold for Z loops and the corresponding m anyons.
- The loop operation is like an interference measurement, in which a beam of particles split and merge to reveal the statistical phase associated with particles inside the loop.

Noncontracting Loops

- Define, for noncontracting loop c, c' ,

$$|\psi\rangle = Z(c) |\psi_{gs}\rangle,$$

$$|\psi'\rangle = X(c') |\psi_{gs}\rangle.$$



- They are degenerate with the ground state, as

$$[H, Z(c)] = [H, X(c')] = 0.$$

- Unlike for contracting loops, $|\psi\rangle = |\psi'\rangle = |\psi_{gs}\rangle$ cannot be correct (i.e., degenerate ground states exist), because

$$Z(c) |\psi'\rangle = Z(c) X(c') |\psi_{gs}\rangle = -X(c') Z(c) |\psi_{gs}\rangle = -X(c') |\psi\rangle.$$

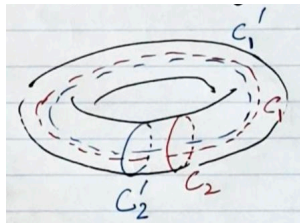
- Such states are invariant under continuous deformations of the anyonic loops; only four orthogonal states can exist.
- Noncontracting loops define logical operations. For example, we can encode two (logical) qubits by

$$|00\rangle = |\psi_{gs}\rangle ,$$

$$|10\rangle = X(c'_1) |\psi_{gs}\rangle ,$$

$$|01\rangle = X(c'_2) |\psi_{gs}\rangle ,$$

$$|11\rangle = X(c'_2) |10\rangle = X(c'_1) |01\rangle .$$



- Similar logical operators can be defined for $Z(c_1)$ and $Z(c_2)$.

Connection to Quantum Error Correction

- Kitaev's toric code model is a quantum model with local stabilizers as interaction.
 - The ground state manifold can be identified as the **encoding space**. The ground-state degeneracy is robust against local perturbation.
 - Excitations are anyons. The global wave function acquires a phase factor when one anyon moves around another. The creation (of anyon pairs) and motion of anyons correspond to **errors** on the physical level. Error syndromes can be tracked by frequent stabilizer measurements. Efficient error extraction (on classical computers) is needed.
 - One can operate on the ground state by creating a pairs of anyons, moving one around the torus, and annihilating with the other. The resulting string operation is a **logical operator**.

Summary

- Modern codes allows fault-tolerant quantum computation as long as the noise is below a certain constant threshold.
 - In surface codes, one can monitor the occurrences of errors by repeated stabilizer measurements. If the errors are sufficiently rare, and they can be undone in classical control software.
- In a toric code/surface code, encoding space can be identified as the ground state manifold, while errors correspond to anyonic excitations. Logical operators are global operations that move an encoded state within the ground state manifold.

References

- A. Yu. Kitaev, *Fault-tolerant quantum computation by anyons*, arXiv:quant-ph/9707021, published as Annals Phys. 303 (2003) 2-30.
- J. K. Pachos, Introduction to Topological Quantum Computation (Cambridge University Press, 2012).